

Profile

	I'm an independent AI systems designer working on epistemic infrastructure — the systems that shape how meaning, agency, and cognition get structured when humans and machines think together. Current work concentrates on persistent AI memory, knowledge graphs, and agent architectures for human–machine and inter-agent collaboration. My practice spans more than two decades across systems engineering, human/ computer interaction design (HCI), full-stack development, and design, and has focused over the past decade on knowledge systems, retrieval architectures, and spatial interfaces. Several earlier projects pinpointed architectural patterns that later entered wider adoption — among them the vector–graph retrieval approach now widely known as GraphRAG (ECCHR Explore, 2020–2021), spatial collaboration through proxemic interaction (Wonder, 2021–2022), and visual no-code orchestration of AI workflows (Hunch, 2023). Current independent research comprises two lines of work: RAGE (Recursive Agentic Graph Embeddings), a research testbed studying how continuous dynamics propagate over co-evolving graph substrates; and Recurse, a frame-based knowledge graph in which typing emerges from use and frames carry their own traversal instructions.
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Institutional Affiliations

Period	Role	Organization
2026 — present	Advisor, STIP Compass Technology Sounding Board	OECD Science, Technology and Innovation Policy division
2025 — present	Affiliated Researcher	Atlas Research Group
2024 — present	Consultant, Independent AI Engineer	OECD Science, Technology and Innovation Policy division

Professional Experience

Period	Role	Organization	Description
2026 — present	Advisor, STIP Compass Technology Sounding Board	OECD	Standing advisory body on emerging-technology questions for the STIP division. Advise on architecture, data processes, and new modes of interaction with STIP Compass data. Focus: agentic systems, knowledge infrastructure, and agentic AI for science-policy work.
2025 — present	Founder, Lead Engineer, Systems Designer	worldwire	Agent-navigable interface for a real-time news corpus. A continuous ingestion pipeline performs topic clustering and community detection over articles, producing a live topical landscape that updates as the news cycle evolves. Agents are enabled to traverse this landscape to surface, contextualize, and follow stories over time, rather than retrieving isolated articles in response to point queries.

2025 – present	Founder, Lead Engineer, Systems Designer	masked.chat	Privacy-preserving AI chat application that detects and redacts personally identifying information in real time, before content reaches the model provider. Investigates how AI-mediated communication can maintain epistemic integrity and user sovereignty under adversarial data-leakage conditions. End-to-end design and engineering: detection pipeline, redaction model, interaction patterns for surfacing what was redacted and why.
2024 – present	AI Systems Designer, Software Engineer	RAGE	RAGE (Recursive Agentic Graph Embeddings) is a research implementation for studying how continuous dynamics — attention, activation, decay, coupling, and phase signatures — propagate over a co-evolving graph substrate whose typing and topology change under those same dynamics. Investigates what stable measurement surfaces can be defined on a substrate that co-evolves with its activity.
2024 – present	Founder, Lead Engineer, Systems Designer	Recurse	Frame-based knowledge graph for navigating and connecting personal research materials — documents, notes, and concepts — at every scale of granularity. Built on frames (after Minsky 1975 and Fillmore 1982): typed, recursively composable semantic units with named slots, defaults, and attached procedures, composing from individual concepts to entire workspaces. Two architectural commitments distinguish the system — typing and structure are emergent from materials and from use rather than declared top-down via a fixed ontology, and frames carry their own traversal instructions, encoding how they can be navigated and combined alongside what they contain.
2024 – present	AI Engineer	OECD CompassGPT	Policy-research agent in active use by analysts at the OECD's Science, Technology and Innovation Policy division. Provides retrieval-augmented access to STIP Compass survey data and the surrounding policy literature, with provenance preserved through retrieval and synthesis steps so that claims remain traceable to their source. Designed, built, and maintained as an independent consultant to the division.
2024 – present	AI Engineer	OECD STIP survey	Data-collection and survey-management infrastructure for the cross-country STIP survey of OECD member countries. Extends the existing collection pipeline with agentic data-collection capabilities, allowing the division to gather, validate, and integrate policy data at higher coverage and lower reporting burden.
2023	Senior HCI Designer	Hunch	Contributed early interaction design and conceptual exploration to a visual programming interface for AI workflow orchestration — chaining models, managing context, and inspecting intermediate outputs. One of the first tools to approach AI model composition as a spatial no-code canvas, a pattern that has since become a de-facto standard in tools such as n8n.
2021 – 2022	Product Lead, Systems Designer, Software Engineer	Tweetscape	Led design and development of a graph-based social network analysis interface funded by the Roote Foundation. Wrote architecture specifications, directed product strategy and led the engineering team. The system gave users agency over their Twitter feed recommendation algorithm through network analysis, spatial clustering, and topic modeling.
2021 – 2022	Senior Interaction Designer	Wonder	Spatial video-conferencing platform supporting thousands of concurrent users. Translated the 2021 brand redesign into a working UI design system — token system (color, spacing, typography), foundational component patterns, accessibility standards — and authored a set of Kinetic Principles for spatial interactions.
2020 – 2021	Design Lead and Data Scientist	ECCHR Explore	Led an assembled a team for the European Center for Constitutional and Human Rights (ECCHR) to turn legal podcast episodes into interactive maps of ideas and their inter-connections. Early production implementation of GraphRAG-style retrieval combining vector search with graph traversal over a legal/discursive corpus.
2020 – 2021	Project Lead	Metasphere/ CORD-19 explorer	Prototype of an embedding-space topic explorer built over the CORD-19 scientific-literature dataset during the COVID-19 period. Mapped inter-paper relationships, producing a navigable 3D-terrain of the live scientific corpus and letting researchers move between coarse thematic structure and individual papers without losing orientation. An early applied test of embedding-space navigation as a research-reading interface.

2017 – 2019	Senior HCI Designer	Metasphere	Self-directed research project on spatial knowledge representation: rendering unstructured document corpora as navigable terrains through topic modeling, embedding spaces, and rule-based layout. Built end to end — the NLP backend (topic-modeling and embedding pipelines, similarity and clustering analysis) and the visual primitives (cartographic metaphors, interaction patterns, navigation models). Origin point of the research line that runs through ECCHR Explore, Tweetscape, Recurse, and the present RAGE work; the first sustained attempt in this practice to treat knowledge architecture and retrieval as a coupled design-and-engineering problem.
2019 – 2021	Co-founder & Managing Partner	Sunsten	Strategic-technology consultancy for decision-makers navigating emerging-technology landscapes. Mapped the early web3/blockchain project ecosystem using NLP over whitepapers, project documentation, and surrounding discourse, turning a hyped and opaque field into a legible terrain for strategic decisions. Delivered the resulting analyses through tooling and structured workshops.
2018 – 2022	Design Director	Ape Unit	Led design across blockchain and web3 projects at a digital agency. Built spatial interfaces, worked with NLP and embedding visualizations, connected design research with applied development. Notable engagement: æternity education platform.
2017 – 2019	Principal Designer & Researcher	Metasphere	Self-directed research project on spatial knowledge representation: turning unstructured document corpora into navigable terrains using topic modeling, embedding spaces, and rule-based layout. Built both the NLP backend (topic-modeling and embedding pipelines, similarity and clustering analysis) and the visual primitives (cartographic metaphors, interaction patterns, navigation models) end-to-end. Metasphere is the origin point of the research line that runs through ECCHR Explore, Tweetscape, Recurse, and the present RAGE work — the first sustained effort to treat knowledge architecture and retrieval as a coupled design-and-engineering problem.
2014 – 2016	Lecturer, Information Design & Data Visualization	Design Factory International	Weekly lectures and workshops on information design, data visualization, emerging technologies (AI, VR), and mapping techniques. Counseled student projects on knowledge exploration and visualization.
2010 – 2016	Design Director	hinterhoflinks	Networked design studio for cultural-institution and corporate clients. Core practice focused on generative and procedural design and the construction of design systems — rule-based visual systems, parametric corporate-identity frameworks, and code-driven generation of assets at scale. Conventional corporate-design and typesetting work (including extensive annual-report production) ran alongside and sustained the studio while the systems practice deepened.
2009 – 2010	Brand Designer & Partner	formvermittlung	Co-founded a corporate-design agency after graduating in Communications Design. Brand strategy, design systems, and early experiments in generative branding.
2004 – 2007	Frontend Developer	freistil Advertising Agency	Frontend development in the early days of modern web development. PHP, SQL, early CSS/JS.
2000 – 2006	Systems Administrator/ Penetration Tester	webrelations ISP	Began professional systems work as a teenager through the Chaos Computer Club community. Proxy services and caching, mail-server hardening, offsite-backup strategies, Unix administration, Perl.

Education

Period	Qualification	Institution
2006 – 2009	Communication Design (Diploma)	Design Factory International, Hamburg, Germany

Areas of Expertise

Area	Level	Years
Knowledge architectures & retrieval systems	Expert	10+
Information architecture & interface design	Expert	10+
Spatial interfaces & knowledge cartography	Expert	10+
Systems architecture & infrastructure	Advanced	20+
Frontend / UI development	Advanced	20+
Data visualization	Advanced	10+
Natural Language Processing Pipelines	Advanced	8
Graph databases & backend	Advanced	8
AI agents & orchestration	Advanced	5

Languages

German	Native
English	full professional proficiency